9 The circumference of circle **B** is 90% of the circumference of circle **A**.

(a) Find the ratio of the area of circle **A** to the area of circle **B**.

(a) Circumference scale B : A = 0.9 : 1

Area scale B : A = $0.9^2 : 1^2 = 81 : 100$

Hence area A : B = 100 : 81

1 mark
Square the length scale.

1 mark
100 : 81.

100 : 81

(2)

Square **E** has sides of length e cm.
Square **F** has sides of length f cm.

The area of square **E** is 44% greater than the area of square **F**.

(b) Work out the ratio $e : f$

(b) Area E : F = 144 : 100

1 mark
Use area multiplier 1.44.

Side E : F = $\sqrt{144} : \sqrt{100} = 12 : 10$

e : f = 6 : 5

1 mark
Take square roots: 6 : 5.

6 : 5

(2)

(Total for Question 9 is 4 marks)